INCH-POUND

MIL-D-44391(GL)
29 December 1989

MILITARY SPECIFICATION

DESSERT BARS, RATION LIGHTWEIGHT, 30-DAY

This specification is approved for use by the Natick Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers dessert bars for use by the Department of Defense as a component of operational rations.

1.2 Classification. The dessert bars shall be of the following types as specified (see 6.1).

Type I - Apple cinnamon dessert bar

Type II - Blueberry dessert bar

Type III - Chocolate chip dessert bar

Type IV - Chocolate halva dessert bar

Type V - Graham dessert bar

Type VI - Pecan dessert bar

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 8940

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.1).

SPECIFICATIONS

FEDERAL

L-P-378	-	Plastic Sheet and Strip, Thin Gauge, Polyolefin
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-C-843	-	Cushioning Material, Cellulosic
PPP-C-1752	-	Cushioning Material, Packaging, Unicellular Polyethylene Foam, Flexible
PPP-C-1797	-	Cushioning Material, Resilient, Low Density, Unicellular, Polypropylene Foam

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

ENVIRONMENTAL PROTECTION AGENCY (EPA)

National Primary Drinking Water Regulations

(Copies are available from the Office of Drinking Water, Environmental Protection Agency, WH550D, 401 M Street, S.W., Washington, DC 20460.)

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U.S. DEPARTMENT OF AGRICULTURE (USDA)

Regulations Governing the Inspection of Eggs and Egg Products (7 CFR Part 59)

(Copies are available from the Poultry Division, Agricultural Marketing Service, U.S. Department of Agriculture, Room 3944, South Building, P.O. Box 96456, Washington, DC 20090-6456).

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES (HHS)

Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder (21 CFR Parts 1-199)

(Copies are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.1).

AMERICAN ASSOCIATION OF CEREAL CHEMISTS (AACC)

Approved Methods of the American Association of Cereal Chemists

(Application for copies should be addressed to the American Association of Cereal Chemists, 3340 Pilot Knob Road, St. Paul, MN 55121.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

F 88 - Seal Strength of Flexible Barrier Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS (AOAC)

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, 2200 Wilson Boulevard, Suite 400-CD, Arlington, VA 22201-3301.)

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NATIONAL ACADEMY OF SCIENCES

Food Chemicals Codex

(Application for copies should be addressed to the National Academy Press, 2101 Constitution Avenue, N.W., Washington, DC 20418.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.1), a sample shall be subjected to first article inspection (see 6.2) in accordance with 4.4.

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent or insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

3.2.1 Chocolate drops. Chocolate drops shall be semi-sweet chocolate, free of dextrose, and shall contain not less than 28 percent fat. The chocolate drops shall have a count per pound of not less than 1800 nor more than 2200.

3.2.2 Cinnamon, ground. Ground cinnamon shall be prepared from the dried bark of Cinnamomum burmanii Blume commonly known as "Korintz Cinnamon". The cinnamon shall contain not less than 1.5 mL of volatile oil per 100 grams of ground cinnamon and shall be of such size that not less than 95 percent shall pass through a U.S. Standard No. 60 sieve.

3.2.3 Cheese, cream, dehydrated. The dehydrated cream cheese shall be derived from fresh cream cheese complying with the FDA Standard of Identity for Cream Cheese (21 CFR 133.133). The dehydrated cheese shall be white to light cream in color, uniform in appearance and composition, and shall possess a characteristic pleasing and desirable flavor. The dehydrated cheese shall be free from specks and burnt particles. The dehydrated cheese shall contain not more than 3.0 percent moisture and not less than 65.0 percent milkfat.

3.2.4 Cream, sweet, spray-dried. Dry sweet cream shall comply with the FDA Standard of Identity for Dry Cream (21 CFR 131.149). The spray-dried sweet cream shall be of an off-white to cream color and have an odor and flavor characteristic of sweet cream. It shall contain 60 to 74 percent, by weight, of milkfat and shall contain no added nutritive sweeteners. The moisture content shall be not greater than 2.0 percent. The spray-dried sweet cream shall be free

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from lumps that do not fall apart under light pressure, and 100 percent of the product shall pass through a U.S. Standard No. 12 sieve. It shall contain lecithin and an antioxidant to improve storage stability.

3.2.5 Whole eggs, liquid or frozen. Whole eggs may be liquid or frozen and shall have been processed and labeled in accordance with the Regulations Governing the Inspection of Eggs and Egg Products. The whole eggs shall be egg whites and egg yolks in their natural proportions as broken directly from the shell eggs as evidenced by a USDA Egg Products Inspection Certificate. For liquid whole eggs, the certificate shall state the date and time of pasteurization. Liquid whole eggs shall be held at a temperature of 40°F or lower and shall be held for not more than 72 hours from the time of pasteurization until the start of formulation of the product in which the liquid whole eggs are used. Frozen whole eggs shall be held at 10°F or lower and used within 120 days from the date of certification. The whole eggs shall be free from off-odors, off-flavors, or foreign odors and flavors, such as metallic, musty, sour, fruity, or sulfide-like, and shall be free from foreign materials.

3.2.6 Flavoring, vanilla, imitation. The imitation vanilla flavoring shall be a single strength imitation of vanilla extract. It shall possess a flavor and odor typical of vanilla flavoring.

3.2.7 Flour, wheat. Wheat flour shall be free-flowing, enriched, general purpose white (bleached).

3.2.8 Glycerine, food grade. Food grade glycerine shall comply with the requirements of the Food Chemicals Codex.

3.2.9 Graham cracker crumb. The graham cracker crumb shall be a crumb from commercial graham crackers containing enriched flour. The graham cracker crumb shall have an odor, color, and flavor characteristic of graham crackers. The graham cracker crumb shall have a maximum moisture content of 6 percent. It shall also have a protein content in the range of 6.5 to 8.5 percent, and a fat content in the range of 7.5 to 10.0 percent. The particle size of the graham cracker crumb shall be such that, by weight, zero percent is retained on a U.S. Standard No. 4 sieve, and a maximum of 10 percent passes through a U.S. Standard No. 80 sieve.

3.2.10 Halva, chocolate-flavored. The chocolate-flavored halva shall consist of crushed sesame, glucose, sugar, and partially hydrogenated vegetable oil (cottonseed and soy bean), cocoa, and chocolate. The halva shall have a fat content in the range of 35.7 to 43.7 percent, a protein content in the range of 11.6 to 14.2 percent, and a maximum moisture content of 3.7 percent. The halva shall be of a medium brown color, and shall possess a typical odor, flavor, and consistency (see 6.4).

3.2.11 Juice, lemon. Lemon juice may be canned or frozen, of a single strength concentration, or concentrated lemon juice, diluted to single strength equivalent. The lemon juice shall be prepared from unfermented juice obtained from fresh, sound, mature lemons of one or more of the high acid varieties.

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3.2.12 Margarine. Margarine shall comply with the FDA Standard of Identity for Margarine (21 CFR 166.110). The margarine shall be manufactured from corn oil and shall possess a fine and pleasing flavor. The body shall be smooth, firm and homogeneous. The product shall possess a uniform medium yellow color. The margarine shall be salted.

3.2.13 Monoglycerides, distilled. Distilled monoglycerides shall be produced from refined hydrogenated soybean oil. It shall be a fine powder, with not more than 5 percent remaining on a U.S. Standard No. 100 sieve, and no more than 10 percent remaining on a U.S. Standard No. 200 sieve. The distilled monoglycerides shall have a minimum monoester content of 90 percent, a maximum iodine value of 5, a free fatty acid content of no more than 1.5 percent, a free glycerol content of no more than 1.0 percent, and a melting point of approximately 162°F (see 6.5).

3.2.14 Nuts, formulated, wheat-based, apple-cinnamon flavored. Formulated wheat-based nuts shall consist of a mixture of hydrogenated soybean oil, sugar, wheat germ, starch, sodium caseinate, apple powder, natural apple flavor, and cinnamon. The formulated apple-cinnamon flavored wheat-based nut shall have an odor and flavor characteristic of apple and cinnamon. The nut shall be tan in color. The moisture content shall have a range from 2.0 to 3.0 percent. The formulated wheat-based nut shall also have a protein content in the range from 7.7 to 9.9 percent, and a fat content in the range from 48.0 to 53.0 percent. The apple-cinnamon flavored wheat-based nut shall have a uniform cut size of 1/4 inch ± 1/16 inch (see 6.6).

3.2.15 Nuts, formulated, wheat-based, blueberry-flavored. Formulated wheat-based nuts shall consist of a mixture of hydrogenated soybean oil, sugar, wheat germ, starch, sodium caseinate, soy protein, natural blueberry flavor, and artificial color. The formulated blueberry-flavored wheat-based nut shall have an odor and flavor characteristic of natural blueberries. The nut shall be blue in color. The moisture content shall have a range from 1.5 to 2.5 percent. The formulated wheat-based nut shall have a protein content that has a range from 8.0 to 10.0 percent, and a fat content in the range from 43.0 to 49.0 percent. The blueberry-flavored formulated wheat-based nut shall have a small piece size, with the following granulation percentages:

Screen size	Product remaining on screen (percent by weight)
5/16	trace
1/4	5 ± 3 percent
3/16	65 ± 10 percent
1/8	25 ± 10 percent
1/8	5 percent maximum may pass through the 1/8 inch mesh screen

(see 6.6)

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3.2.16 Nuts, formulated, wheat-based, English walnut and pecan, unsalted. Formulated wheat-based nuts, unsalted shall consist of a mixture of hydrogenated soybean oil, wheat germ, sugar, sodium caseinate, soy protein, either artificial English walnut or pecan flavors, as applicable, and artificial coloring. The formulated wheat-based nuts shall have an odor and flavor characteristic of English walnuts or pecans. The color of the nut shall be light tan or beige in the interior with an exterior color that is similar to the fresh nut. The moisture content shall not be greater than 3.0 percent. The formulated wheat-based nut shall have a protein content in the range of 8.5 to 14.0 percent, and a fat content ranging from 59.0 to 65.0 percent. A small size nut shall be used for the English walnut flavored wheat-based nut. For the pecan a large nut size shall be used. Granulation percentages, as determined by using round hole screen are for each size as follows:

	Product remaining on screen (percent by weight)	
Screen size	Small size	Large size
5/16	trace	85 ± 10 percent
1/4	5 ± 3 percent	15 ± 5 percent
3/16	65 ± 10 percent	5 percent max
1/8	25 ± 10 percent	trace
1/8	5 percent maximum may pass through the 1/8 inch mesh screen	

(see 6.6)

3.2.17 Salt. Salt shall be iodized, white, refined sodium chloride with or without anticaking agents and shall comply with the purity standards for sodium chloride of the Food Chemicals Codex.

3.2.18 Sugar, white, granulated. Sugar shall be white refined, granulated cane or beet sugar, or a combination thereof.

3.2.19 Syrup solids, corn, powdered. Powdered corn syrup solids shall be produced from corn syrup having a dextrose equivalent (DE) of 42. The corn syrup solids shall be bland in flavor and shall have a moisture content of not more than 5.0 percent, an SO_2 content of not more than 40 ppm, and a pH of 4.7. The corn syrup solids shall contain a carbohydrate composition of approximately 20 percent monosaccharides, 14 percent disaccharides, 11 percent trisaccharides, and 55 percent tetrasaccharides and higher carbohydrates. One hundred percent of the powdered solids shall pass through a U.S. Standard No. 40 sieve and not less than 90 percent shall pass through a U.S. Standard No. 100 sieve.

3.2.20 Water. Water used for formulation and sanitizing shall conform to the National Primary Drinking Water Regulations.

3.3 Preparation and processing. Processing shall be on a continuous basis.

3.3.1 Preparation of matrix. A separate matrix shall be formulated and prepared for type I, II, III, IV, and VI products as follows:

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3.3.1.1 Preparation of cinnamon and plain matrices .

Cinnamon matrix for type I

Ingredients	Percent by weight
Flour, wheat	48.8
Margarine	35.9
Sugar, white, granulated	10.0
Cinnamon	3.0
Juice, lemon	2.0
Salt	0.3

Plain matrix for types II and VI.

Ingredients	Percent by weight
Flour, wheat	51.79
Margarine	26.38
Sugar, white, granulated	10.78
Whole eggs, liquid or frozen 1/	8.80
Juice, lemon	1.96
Salt	0.29

1/ Frozen whole eggs shall be tempered/thawed and held at an internal temperature of 28° to 40°F for not more than 24 hours prior to product preparation.

The cinnamon and plain matrices shall be prepared as follows:

a. All ingredients shall be mixed together in a suitable mixer until margarine is in chunks or pieces approximately 1/2 inch in diameter.

b. The mixture shall be poured and flattened, if necessary, on a baking sheet, to a thickness between 1/4 inch and 1/2 inch.

c. The matrix shall be baked at 250°F, until moisture is less than 2.0 percent. There shall be no evidence of excessive browning, burning, or scorching. The color of the cinnamon matrix shall be light to medium tan, and the odor and flavor shall be that of a baked cinnamon-flavored pie crust. The color of the plain matrix shall be off-white to light tan, and the odor and flavor shall be that of a baked pie crust.

d. The prepared matrix shall be stored in a vacuum sealed container until ready to use. Matrix may be held for up to 30 days before forming into bars provided the matrix is held at a temperature below 75°F.

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3.3.1.2 Preparation of cookie matrix for types III and IV. 1/

Ingredients	Percent by weight
Flour, wheat	48.0
Margarine	25.0
Sugar, white, granulated	21.0
Water	5.0
Salt	0.6
Imitation flavoring, vanilla	0.4

The cookie matrix shall be prepared as follows:

a. Dry ingredients shall be blended in a Hobart type mixer.

b. The margarine shall be added and mixed until margarine is in chunks or pieces approximately 1/2 inch in diameter.

c. The water and vanilla shall be blended in until just mixed.

d. The mixture shall be poured and flattened, if necessary, on a baking sheet to a thickness between 1/4 inch and 1/2 inch.

e. The matrix shall be baked at 250°F, until moisture is less than 2.0 percent. There shall be no evidence of excessive browning, burning, or scorching. The color of the matrix shall be light to medium tan. The odor and flavor shall be that of a baked cookie dough.

f. The matrix shall be stored in a vacuum sealed container until ready to use. Matrix may be held for up to 30 days before forming into bars provided the matrix is held at a temperature below 75°F.

1/ Matrix may have to be milled through a 3/4 inch screen to reduce piece size prior to formulation and preparation of dessert bars (see 3.3.2.7).

3.3.2 Dessert bar formulation and preparation. The dessert bars shall be formulated and prepared as follows:

3.3.2.1 Type I - Apple cinnamon dessert bar.

Ingredients	Percent by weight
Nuts, formulated, wheat-based, apple-cinnamon flavored	68.8
Matrix, cinnamon	20.0
Sugar, white, granulated	8.5
Juice, lemon	1.5
Cinnamon	0.7
Glycerine	0.5

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3.3.2.2 Type II - Blueberry dessert bar.

Ingredients	Percent by weight
Nuts, formulated, wheat-based, blueberry flavored	49.6
Matrix, plain	39.6
Sugar, white, granulated	8.0
Glycerine	1.7
Juice, lemon	1.1

3.3.2.3 Type III - Chocolate chip dessert bar.

Ingredients	Percent by weight
Cookie matrix	36.0
Nuts, formulated, wheat-based, English-walnut flavored	33.0
Chocolate drops	31.0

3.3.2.4 Type IV - Chocolate halva dessert bar.

Ingredients	Percent by weight
Cookie matrix	33.00
Halva, chocolate flavored 1/	33.00
Chocolate drops	31.00
Monoglycerides, distilled 1/	2.75
Glycerine 1/	0.25

1/ Heat halva until softened (120°to 140°F), and incorporate glycerine and distilled monoglycerides until well mixed. Cool mixture to 70° to 75°F before adding cookie matrix and chocolate drops.

3.3.2.5 Type V - Graham dessert bar.

Ingredients	Percent by weight
Graham craker crumb	34.00
Cream cheese, dehydrated	30.50
Sugar, white, granulated 1/	24.14
Margarine 1/	11.30
Imitation flavoring, vanilla	0.06

1/ Blend sugar and margarine to a creamy consistency before adding the remaining ingredients.

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3.3.2.6 Type VI - Pecan dessert bar.

Ingredients	Percent by weight
Nuts, formulated, wheat-based, pecan flavored, unsalted	35.0
Matrix, plain	30.0
Cream, sweet, spray-dried	15.0
Sugar, white, granulated	10.0
Syrup solids, corn, powdered, 42 DE	10.0

3.3.2.7 Dessert bar preparation. Mix ingredients for each type bar in a suitable mixer until liquid ingredients are well dispersed, well blended, and no pieces larger than 3/4 by 3/4 inch exist.

3.4 Compression of dessert bars. The dessert mix shall be compressed so as to produce bars complying with the finished product requirements (see 3.6).

(NOTE: Satisfactory product has been compressed at pressures ranging from 300 to 700 psi.)

3.5 Pouch filling and sealing. The compressed bars shall be packaged immediately in accordance with 5.1.

3.6 Finished product requirements. The finished product shall comply with the following requirements:

a. There shall be no foreign material such as, but not limited to, dirt, insect parts, hair, wood, glass, or metal.

b. There shall be no foreign odors or flavors, such as, but not limited to, burnt, scorched, stale, sour, rancid, or moldy.

c. There shall be no color foreign to the product.

d. There shall be no compression streaks or stains.

e. Each bar shall be 1.0 $\pm$ 0.1 inch wide by 3.0 $\pm$ 0.1 inches long by 1.0 $\pm$ 0.1 inch thick.

f. The weight of each bar shall be not less than 50 grams nor greater than 60 grams.

g. No pouch shall contain a bar which has been broken or crumbled (more than three pieces).

h. The moisture content shall be not greater than 5.0 percent.

i. The aerobic plate count shall be not greater than 75,000 per gram in four of five samples and not greater than 150,000 per gram in any sample.

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j. The E. coli count shall be less than 3 per gram (no positives in the standard three tube MPN technique) in four of five samples and not greater than 20 per gram in any sample .

k. The compressed type I apple cinnamon bar shall be light to medium brown in color and the odor and flavor shall be typical of apple and cinnamon.

l. The compressed type II blueberry bar shall be light to medium tan in color with medium blue pieces of flavored formulated nuts. The odor and flavor shall be typical of natural blueberries in a baked dough.

m. The compressed type III chocolate chip bar shall be tan in color with dark brown spots of chocolate chips. The odor and flavor shall be typical of a chocolate chip cookie.

n. The compressed type IV chocolate halva bar shall be medium brown and tan in color and the odor and flavor shall be typical of chocolate halva.

o. The compressed type V graham bar shall be light to medium tan in color. The bar shall possess the odor and flavor of graham and cream cheese.

p. The compressed type VI pecan bar shall be off-white to light tan in color with brown specks. The flavor and odor shall be similar to pecans in a baked dough.

3.6.1 Palatability. The finished product shall be equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance.

3.7 Plant qualification. The product shall be prepared, processed, and packaged in establishments meeting the requirements of Title 21, Code of Federal Regulations, Part 110, "Current Good Manufacturing Practice in Manufacturing, Packing or Holding Human Food", and the plant sanitation requirements of the appropriate Government inspection agency.

3.8 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this specification. The contractor shall ensure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and certification. Product acceptability shall be determined by the USDA. The USDA will determine the degree of inspection necessary to ensure compliance with the requirements of this specification.

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prepared from highly purified food grade Acting". The baking powder shall be p

4.3 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be inspected in accordance with the quality assurance provisions of this specification and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this specification or any appearance or palatability failure shall be cause for rejection of the first article.

4.5 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.5.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.5.1.1 Ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements specified in 3.2 shall be certified by the ingredient supplier or ingredient manufacturer, and compliance shall be verified by examination of pertinent labels, markings, U.S. Grade Certificates, certificates of analyses, or other such valid documents acceptable to the inspection agency. If necessary, each ingredient shall be examined organoleptically or inspected according to generally recognized test methods, such as the standard methods described in the Official Methods of Analysis of the Association of Official Analytical Chemists, and in the Approved Methods of the American Association of Cereal Chemists, to determine conformance to the requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

4.5.1.2 Laminated pouch material certification. A certificate of compliance may be accepted as evidence that the characteristics listed below conform to the specified requirements.

Material requirement	Requirement paragraph	Test procedure
Thickness of films	5.1.1.1.1	As specified in L-P-378, except that a machinist's micrometer may be used provided that its graduations and accuracy conform to the requirements of L-P-378.

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the contract or purchase order, shipping containers shall be marked in

Laminated material identification and construction	5.1.1.1.1 and 5.1.1.2.1	Laboratory evaluation.
Total thickness	5.1.1.2.1	As specified in L-P-378, except that a machinist's micrometer may be used provided that its graduations and accuracy conform to the requirements of L-P-378.

4.5.1.3 Unfilled pouch certification. A certificate of compliance may be accepted as evidence that unfilled pouches conform to the requirements specified in 5.1.1.1.1 and 5.1.1.2.1. When deemed necessary by the USDA, testing of the unfilled preformed pouches for seal strength and integrity and air tightness of manufacturer's seals shall be as specified in 4.5.1.3.1 and 4.5.1.3.2.

4.5.1.3.1 Unfilled preformed pouch seal strength testing. The unfilled pouches shall be tested for seal strength in accordance with ASTM F 88, except that the testing speed shall be 10 or 12 inches per minute. Machines that apply the tensile load to the test specimen by movement of the upper or lower clamp may be used. Test specimens shall be cut 1/2 or 1 inch in width and to a length suitable for proper mounting. The lot size shall be expressed in pouches. The sample unit shall be one unfilled pouch. The sample size shall be the number of pouches indicated by inspection level S-1. Three adjacent specimens shall be cut from each of the three sealed sides of each pouch in the sample. The results shall be reported to the nearest 0.1 pound per inch of width. The average seal strength of each seal shall be calculated by averaging the test results of the three test specimens cut from that seal. Any test specimen failing to meet the individual test specimen seal strength requirement or any seal failing to meet the average seal strength requirement specified in 5.1.1.1.1 or 5.1.1.2.1 shall be classified as a major defect and shall be cause for rejection of the lot.

4.5.1.3.2 Unfilled preformed pouch integrity and air tightness of manufacturer's seals testing. Pouch integrity and air tightness of seals shall be determined by placing rigid material similar in size and shape to the dessert bar(s) into the pouch and sealing under a vacuum of not less than 25 inches of mercury. Not less than 96 hours after vacuum sealing, the pouch shall be examined for vacuum loss in accordance with the procedure in 4.5.4. Any evidence of loss of vacuum shall be classified as a major defect. The lot size shall be expressed in pouches. The sample unit shall be one unfilled pouch. The inspection level shall be I and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 0.65.

4.5.2 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, filling, sealing, and packaging requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation, or of other valid documents shall be cause for rejection of the involved product.

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4.5.2.1 In-process examination at the assembly plant. Bulk-packed dessert bars shall be examined before unit packing into individual pouches. The lot size shall be expressed in shipping containers. The sample shall be the number of shipping containers indicated by inspection level S-3. Two bars shall be randomly selected from each of the shipping containers in the sample. Any bar found to have objectionable taste or odor (for example, scorched, rancid, burnt, stale, sour, or moldy) shall be cause for rejection of the lot.

4.5.3 Filled and sealed pouch examination (applicable to unit packed and bulk packed pouches). The filled and sealed pouches shall be examined for the defects listed in table I. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be I and the AQL, expressed in terms of defects per hundred units, shall be 0.65 for major defects and 4.0 for minor defects.

TABLE I. Filled and sealed pouch defects (applicable to unit packed and bulk packed pouches) 1/

Category		Defect
Major	Minor	
101		Pouch size not as specified 2/
102		Not heat sealed as specified
103		Not clean 3/
104		Tear, hole, or open seal
105		Pouch has foreign odor
106		Evidence of delamination 4/
107		Seal widths not as specified
108		Closure seal located less than 1/8 inch above tear notch 5/
109		Presence of foldover wrinkles, occluded matter, and entrapped moisture or grease that reduces the effective closure seal to less than 1/8 inch 6/
110		Required labeling or marking missing, incorrect, illegible or that smudges
	201	Tear notch missing 5/
	202	Tear notch not located as specified 5/
	203	Depth of tear notch not as specified 5/

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1/ Any evidence of rodent or insect infestation shall be cause for rejection of the lot.

2/ Measurement may be accomplished on unfilled pouches being used for current production lot.

3/ Outer packaging shall be free from foreign matter which is unwholesome, has the potential to cause pouch damage (for example, glass, metal filings, etc.) or generally detracts from the clean appearance of the pouch. The following examples shall not be scored as defects for unclean:

a. Foreign matter which presents no health hazard or potential pouch damage and which can be readily removed by gently shaking the pouch or by gently brushing the pouch with a clean dry cloth.

b. Dried product which affects less than 1/8 of the total surface area of one pouch face (localized and aggregate).

c. Water spots.

4/ Delamination shall be scored as a defect except delamination of outer ply when located in the seal area 1/16 inch or further from food product edge of seal. Pouches exhibiting this type of delamination shall be tested by manually flexing the delaminated area 10 times. The area of delamination shall be held between the thumb and forefinger of each hand with both thumbs and forefingers touching each other. The delamination area shall then be rapidly flexed by rotating both hands in alternating clockwise - counter-clockwise directions. Care shall be exercised when flexing delaminated area near the tear notches to avoid tearing the pouch material. After flexing, the separated outer ply shall be grasped between the thumb and forefinger and gently lifted toward the food product edge of the seal. If the separated area is too small to be held between the thumb and forefinger, a number two stylus shall be inserted into the delaminated area and a gentle lifting force applied against the outer ply. If separation of the outer ply can be made to extend to less than 1/16 inch from the product edge of the seal with no discernible resistance to the gentle lifting, it shall be considered evidence of delamination.

5/ Applicable to unit packed pouches only.

6/ The effective closure seal is defined as any uncontaminated, fusion bonded, continuous path, minimum 1/8 inch wide from side seal to side seal that produces a hermetically sealed pouch.

4.5.4 Pouch vacuum examination. Not less than 96 hours after filling and sealing, the pouches shall be visually examined for conformance to the vacuum requirements in 5.1.1.1.2 and 5.1.1.2.2. The sealed pouch shall continue to exhibit tight adherence to the surface contours of the contents when a pulling force is applied at the center of each side seal. This force shall be applied by

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holding each side seal between thumb and forefinger of each hand, simultaneously exerting a slight pull with both hands. Any evidence of loss of vacuum shall be classified as a major defect. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be I and the AQL, expressed in terms of defects per hundred units, shall be 0.65.

4.5.5 Net weight examination of product when unit packed in pouches as specified in 5.1.1.1. The net weight of the filled and sealed pouches shall be determined by weighing each sample unit on a suitable scale tared with a representative empty pouch. Any individual net weight of less than 50 grams or greater than 60 grams shall be classified as a minor defect. The lot size shall be expressed in pouches. The results shall be reported to the nearest gram. The sample unit shall be one filled and sealed pouch. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

4.5.6 Product examination when unit packed as specified in 5.1.1.1. The finished product shall be examined for the defects listed in table II. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 1.5 for major defects and 6.5 for minor defects.

TABLE II. Product defects (unit pack) 1/ 2/

Category		Defect
Major	Minor	
101		Presence of compression stains or streaks
102		Dimensions of bar not as specified
103		Color of bar not as specified
104		Odor and flavor of bar not as specified
	201	Broken or crumbled bar (more than three pieces) 3/

1/ The presence of foreign material (for example, dirt, insect parts, hair, wood, glass, metal), foreign odor or flavor (for example, burnt, scorched, moldy, rancid, sour, stale), or foreign color shall be cause for rejection of the lot.

2/ Product not equal to or better than the approved preproduction sample in palatability and overall appearance shall be cause for rejection of the lot. (This comparison shall be performed only when deemed necessary by a USDA agent).

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3/ A broken or crumbled bar is a bar which either:

a. Has been broken into more than three pieces, the smallest of which is larger than 1/2 cubic inch in volume; or

b. Has more than 1/8 of the volume of the bar crushed.

4.5.7 Product testing when unit packs are held over 30 days (see 5.1.1.1). When unit packed bars are held over 30 days before final assembly, the storage container shall be tested for conformance to the oxygen content or vacuum requirements of 5.1.1.1. Testing shall be as specified in 4.5.7.1 or 4.5.7.2. The lot size shall be expressed in number of storage containers. The sample unit shall be one sealed container. The inspection level shall be S-2. Any storage container failing to meet the oxygen content or vacuum requirements shall be cause for rejection of the lot.

4.5.7.1 Oxygen content testing. The analysis to determine the percentage of oxygen in the nitrogen atmosphere shall be in accordance with any accepted USDA method, using appropriate methods to adapt to the bulk container in which the bars are stored.

4.5.7.2 Vacuum testing. The vacuum level shall be determined by testing in accordance with any accepted USDA method, using appropriate methods to adapt to the bulk container in which the bars are stored.

4.5.8 Examination for weight of product when bulk packed in pouches as specified in 5.1.1.2. The weight of the dessert bars shall be determined by individually weighing two bars randomly selected from each sample unit. Any individual bar weight of less than 50 grams or greater than 60 grams shall be classified as a minor defect. The lot size shall be expressed in bulk packed pouches. The results shall be reported to the nearest gram. The sample unit shall be one bulk packed pouch. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

4.5.9 Product examination when bulk packed as specified in 5.1.1.2. The finished product shall be examined for defects listed in table III. The lot size shall be expressed in bulk packed pouches. The sample unit shall be two bars selected at random from the bulk packed pouch. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 1.5 for major defects and 6.5 for minor defects.

TABLE III. Product defects (bulk pack) 1/ 2/

Category		Defect
Major	Minor	
101		Presence of compression stains or streaks
102		Dimensions of bars not as specified

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TABLE III. Product defects (bulk pack) 1/ 2/ (cont'd)

Category		Defect
Major	Minor	
103		Color of bars not as specified
104		Odor and flavor of bars not as specified
105		Number of bars not as specified 3/
	201	Broken or crumbled bar(s) (more than three pieces) 4/

1/ The presence of foreign material (for example, dirt, insect parts, hair, wood, glass, metal), foreign odor or flavor (for example, burnt, scorched, moldy, rancid, sour, stale), or foreign color shall be cause for rejection of the lot.

2/ Product not equal to or better than the approved preproduction sample in palatability and overall appearance shall be cause for rejection of the lot. (This comparison shall be performed only when deemed necessary by a USDA agent).

3/ The sample unit shall be the selected bulk packed pouch.

4/ A broken or crumbled bar(s) is a bar which either:

a. Has been broken into more than three pieces, the smallest of which is larger than 1/2 cubic inch in volume; or

b. Has more than 1/8 of the volume of the bar crushed.

4.5.10 Pouch closure seal strength testing. The filled and sealed pouches shall be tested for closure seal strength in accordance with ASTM F 88, except that the testing speed shall be 10 or 12 inches per minute. Machines that apply the tensile load to the test specimen by movement of the upper or lower clamp may be used. Test specimens shall be cut 1/2 inch in width and to a length suitable for proper mounting. Three adjacent specimens shall be cut from the closure seal of each pouch in the sample. The average seal strength of the closure seal shall be calculated by averaging the test results of the three test specimens cut from the seal. The results shall be reported to the nearest 0.1 pound per inch of width. Any test specimen or average seal strength failing to meet the requirements of 5.1.1.1.2 or 5.1.1.2.2 shall be classified as a major defect and shall be cause for rejection of the lot. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be S-1.

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4.5.11 Moisture content testing. The finished product shall be tested for moisture content in accordance with the Official Methods of Analysis of the Association of Official Analytical Chemists; Chapter: Cereal Foods; Section: Total Solids (Moisture, Indirect Method); Method: Vacuum Oven Method except that the drying cycle shall be 16 hours at 70°C under a pressure of not more than 100 mm Hg. The test results shall be reported to the nearest 0.1 percent. Any test failure shall be classified as a major defect and shall be cause for rejection of the lot. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed unit packed pouch or two bars selected at random from bulk packed pouches. The inspection level shall be S-2.

4.5.12 Microbiological testing. Five filled and sealed unit packed pouches or two bars selected from each of five filled and sealed bulk packed pouches shall be selected at random from the lot regardless of lot size. The pouches or two bar units shall be individually tested for microbiological levels in accordance with the Official Methods of Analysis of the Association of Official Analytical Chemists; Chapter: Microbiological Methods; Section: Frozen, Chilled, Precooked, or Prepared Foods - Official First Action. The diluent shall be added to each bar and allowed to stand for 15 minutes before blending the sample. Continue with the procedure as directed in the Official First Action. Any result not conforming to the microbiological requirements in 3.6 shall be classified as a major defect and shall be cause for rejection of the lot.

4.5.13 Shipping container examination. Shipping containers shall be examined for defects in assembly, closure, and reinforcement (when applicable) in accordance with PPP-B-636. In addition, the following defects shall be classified as follows:

Major: National stock number, item description, contract number, or date of pack markings missing, incorrect, or illegible.
Cushioning material missing or not as specified.

Minor: Other required markings missing, incorrect, or illegible.
Number of bars not as specified.
Pouches not packed flat in layers.

5. PACKAGING

5.1 Preservation. Preservation shall be level A.

5.1.1 Level A. The product shall be unit packed in accordance with 5.1.1.1 or bulk packed in accordance with 5.1.1.2.

5.1.1.1 Unit packs. When specified (see 6.1), one dessert bar shall be unit packed in a preformed pouch as specified in 5.1.1.1.1. The unit packed bars shall be stored in a closed container. If bars are held for more than 30 days from date of pack to ration assembly, they shall be protected from oxygen by either holding under a nitrogen atmosphere with 2.0 percent or less oxygen, or

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holding under a vacuum of at least 27 inches of mercury during the period exceeding 30 days. In no case shall the unit packed bars be held for more than 90 days from date of pack to ration assembly.

5.1.1.1.1 Preformed pouch for unit pack. The preformed pouch shall be fabricated from inside to outside 0.004 inch thick (± 0.001 inch) polyolefin film laminated to 0.0005 inch thick (± 0.0001 inch) polyester. The material shall be suitably formulated for food packaging and shall not impart an odor or flavor to the product being packed. The material shall show no evidence of delamination, degradation, or foreign odor when heat sealed or fabricated into pouches. The preformed pouch shall be a flat style pouch having inside dimensions of 2-3/8 inches wide by 5-3/8 inches long (± 1/16 inch), with the 2-3/8 inch dimension being the opening of the pouch. The pouch shall be made by heat sealing three edges with 1/4 inch (± 1/16 inch) wide seals. The heat seals shall be made in a manner that will ensure hermetic seals. The pouch shall maintain its integrity and air tightness of the side and bottom seals when tested as specified in 4.5.1.3.2. The side and bottom seals shall have an average seal strength of not less than 10 pounds per inch and no individual specimen shall have a seal strength of less than 9 pounds per inch when tested as specified in 4.5.1.3.1. A tear notch at least 1/32 inch deep, located 1 to 1-1/4 inches from the top edge of the pouch, shall be made in one or both side seals. The distance between the inside edge of the tear notch and the inside edge of the seal shall be at least 3/16 inch. One side of the open end of the pouch shall be provided with an extended lip, extending 1/8 inch (± 1/16 inch) to facilitate opening and filling. Tear notch location shall be measured from the top of the pouch, excluding the extended lip.

5.1.1.1.2 Filling and sealing of unit pack pouches. One dessert bar shall be placed into a preformed pouch as specified in 5.1.1.1.1. The filled pouch shall then be hermetically sealed by evacuating the air within a sealing chamber under a vacuum of not less than 25 inches of mercury, and applying a continuous heat seal of not less than 1/8 inch wide. The closure seal shall be free of foldover wrinkles, delamination, occluded matter, and entrapped moisture or grease. The closure seal shall be located not less than 1/8 inch above the tear notch. The average closure seal strength shall be not less than 10 pounds per inch and no individual test specimen shall be less than 9 pounds per inch when tested as specified in 4.5.10. The filled and sealed pouch shall show no loss of vacuum when examined as specified in 4.5.4.

5.1.1.2 Bulk packs. When specified (see 6.1), thirty dessert bars of one type shall be bulk packed in a pouch as specified in 5.1.1.2.1. The bulk packed bars shall be held for not more than 180 days from the date of pack to ration assembly.

5.1.1.2.1 Preformed pouch for bulk pack. The preformed pouch shall be fabricated from a hot fill type, minimum 0.006 inch thick, 3-ply laminate from inside to outside of polyolefin/aluminum foil/polyester material. The material shall be suitably formulated for food packaging and shall not impart an odor or flavor to the product being packed. The material shall show no evidence of delamination,

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degradation, or foreign odor when heat sealed or fabricated into pouches. The preformed pouch shall be a flat style pouch having outside dimensions of 12 inches wide by 15 inches long (± 1/8 inch), with the 12 inch dimension being the opening of the pouch. The pouch shall be made by heat sealing three edges with 3/8 inch (± 1/8 inch) wide seals. The heat seals shall be made in a manner that will ensure hermetic seals. The pouch shall maintain its integrity and air tightness of the side and bottom seals when tested as specified in 4.5.1.3.2. The side and bottom seals shall have an average seal strength of not less than 10 pounds per inch and no individual specimen shall have a seal strength of less than 9 pounds per inch when tested as specified in 4.5.1.3.1.

5.1.1.2.2 Filling and sealing of bulk pack pouches. Thirty dessert bars of one type shall be single tier packed into the bulk pack pouch. The filled pouch shall then be hermetically sealed by evacuating the air within a sealing chamber under a vacuum of not less than 25 inches of mercury, and applying a continuous heat seal of not less than 1/8 inch wide. The closure seal shall be free of foldover wrinkles, delamination, occluded matter, and entrapped moisture or grease. The average closure seal strength shall be not less than 10 pounds per inch and no individual test specimen shall be less than 9 pounds per inch when tested as specified in 4.5.10. The filled and sealed pouch shall show no loss of vacuum when examined as specified in 4.5.4.

5.2 Packing for shipment to ration assembler. Two hundred and ten packaged bars, preserved as specified in 5.1, shall be packed flat in layers in a snug-fitting fiberboard shipping container conforming to style RSC-L, type CF, variety SW, grade 275, class domestic of PPP-B-636. Cushioning material not less than 1/8 inch thick, conforming to type II, class B of PPP-C-843 or cushioning material not less than 1/16 inch thick, conforming to type I of PPP-C-1797 or cushioning material conforming to type VII, class 1 of PPP-C-1752 shall be placed between each of the layers and at the bottom and top of the box to minimize movement of the individual packs.

5.3 Labeling and marking.

5.3.1 Unit pack pouches. Unit pack pouches shall be labeled in a permanent, contrasting color directly on the packaging material in a manner that does not damage the pouches, with a food compatible ink which is free of carcinogenic elements or ingredients. No marking shall be closer than 1/16 inch to any seal. Pressure-sensitive labels may be used instead of marking directly on the packaging material. Labels shall be of a water-resistant grade of paper or film, flat-back, coated on one side with pressure-sensitive adhesive. The texture of the material shall permit flexibility. Labels shall be suitable for printing or writing on with ink without feathering or spreading. The adhesive shall be of a pressure-sensitive permanent type. The adhesive shall be water-insoluble, homogenous, and shall be coated in a smooth layer on one side of the label. The adhesive shall require no solvent, heat, or other preparation prior to application. Adhesive shall be of a type that will adhere to the pouch surface under high or low temperatures. Dimensions of the label shall not be larger than the dimensions of the surface of the product where the label is applied on the

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package. Ink and adhesive used must be compatible for use with food products and shall be free of carcinogenic elements or ingredients. Letter size shall be not less than 1/16 inch high. The following information must be included:

Item Name
Date of production/lot number (Julian date) 1/
Name of contractor

1/ The lot number shall be expressed as a four digit Julian code. The first digit shall indicate the year of production and the next three digits shall indicate the day of the year (Example, March 19, 1987 would be coded as 7078). The Julian code shall represent the day the product was packaged and processed. Sub-lotting (when used) shall be represented by an alpha character immediately following the four digit Julian code.

5.3.2 Bulk pack pouches. Bulk pack pouches shall be marked as specified in 5.3.1, except that the labels shall include the following additional information:

Number of bars
Contract number
Name, address and zip code of contractor

5.3.3 Marking of shipping containers. Shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Type (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When a first article is required (see 3.1, 4.4, and 6.2).
e. Provisions for approved preproduction samples (see 3.6.1 and 6.2).
f. Whether bars are to be unit packed or bulk packed (see 5.1.1.1 and 5.1.1.2).

6.2 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

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6.3 Appropriate level of pack. Based on the conditions known or expected to be encountered during shipment, handling, and storage of the specific item being procured, the procuring activity should select the appropriate level of pack in accordance with the criteria established in AR 700-15/NAVSUPINST 4030.28/AFR 71-6/MCO 4030.33A/DLAR 4145.7.

6.4 Halva, chocolate-flavored. It has been found that product manufactured by Joyva, Inc., meets the requirements of 3.2.10 and performs satisfactorily in this product.

6.5 Monoglycerides, distilled. It has been found that "Diomodan PV 300K" manufactured by Grinsted Inc., meets the requirements of 3.2.13 and performs satisfactorily in this product.

6.6 Nuts, formulated, wheat-based. It has been found that product manufactured by Anacon Foods, Inc. meets the requirements of 3.2.14, 3.2.15, and 3.2.16, and performs satisfactorily in this product.

6.7 Subject term (key word) listing.

Combat field feeding
Food processing
Operational ration

Custodian:

Army - GL

Review activities:

Army - MD, QM
Navy - MC, SA
Air Force - 50
DP - SS

Preparing activity:

Army - GL

(Project 8940-A670)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-D-44391(GL)	Dessert Bars, Ration Lightweight, 30-Day

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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